



DEPARTMENT OF ENGINEERING CYBERNETICS

TTK4900

MASTER THESIS

Adaptive Filtering of Blood Velocity Signals During Running

Author:

Anders Haarberg Eriksen

Date: 28th August 2026

Abstract

Acknowledgments

Contents

1	Introduction	1
2	Theoretical background	2
2.1	Ultrasound	2
2.1.1	Transducer	2
2.1.2	Reflection and Scattering	2
2.1.3	Attenuation	2
2.1.4	Time of Flight	2
2.1.5	Noise sources	2
2.2	Pulsed-Wave Doppler	2
2.2.1	Decimation	3
2.2.2	Clutter Filter	3
2.3	IMU	3
2.3.1	Sensor Fusion	3
3	Method	5
4	Results	6
5	Discussion	7
6	Conclusion	8

List of Figures

2.1	Ideal emitted electrical waveform for Pulsed-Wave Doppler (PW Doppler), with annotations. Note that the time axis is broken to show the pulse repetition period.	3
-----	--	---

List of Tables

Chapter 1

Introduction

Chapter 2

Theoretical background

2.1 Ultrasound

Ultrasound is defined as sound waves with frequencies above the upper limit of human hearing, which is typically around 20 kHz.

2.1.1 Transducer

2.1.2 Reflection and Scattering

2.1.3 Attenuation

2.1.4 Time of Flight

2.1.5 Noise sources

2.2 Pulsed-Wave Doppler

In the present system, PW Doppler is used to measure the blood flow velocity. It works by emitting a series of high frequency sound pulses into the tissue at a constant Pulse Repetition Frequency (PRF). Each pulse consists of a burst of sound waves with a pulse center frequency f_c with N_{cycles} cycles. The pulse length T is then given by $T = N_{cycles}/f_c$. After a pulse is emitted, the system records the returning echoes. This is commonly referred to as the *fast time* signal. The fast time signal is sampled with a sampling rate $f_{s,FT}$

Parameter	Symbol	Value
Pulse center frequency	f_c	8 MHz
Number of cycles	N_{cycles}	12
Pulse length	T	1.5 μs
Pulse repetition frequency	PRF	20 kHz
Speed of sound in tissue	c	1540 m s^{-1}

Ideal Electrical Waveform for PW Doppler

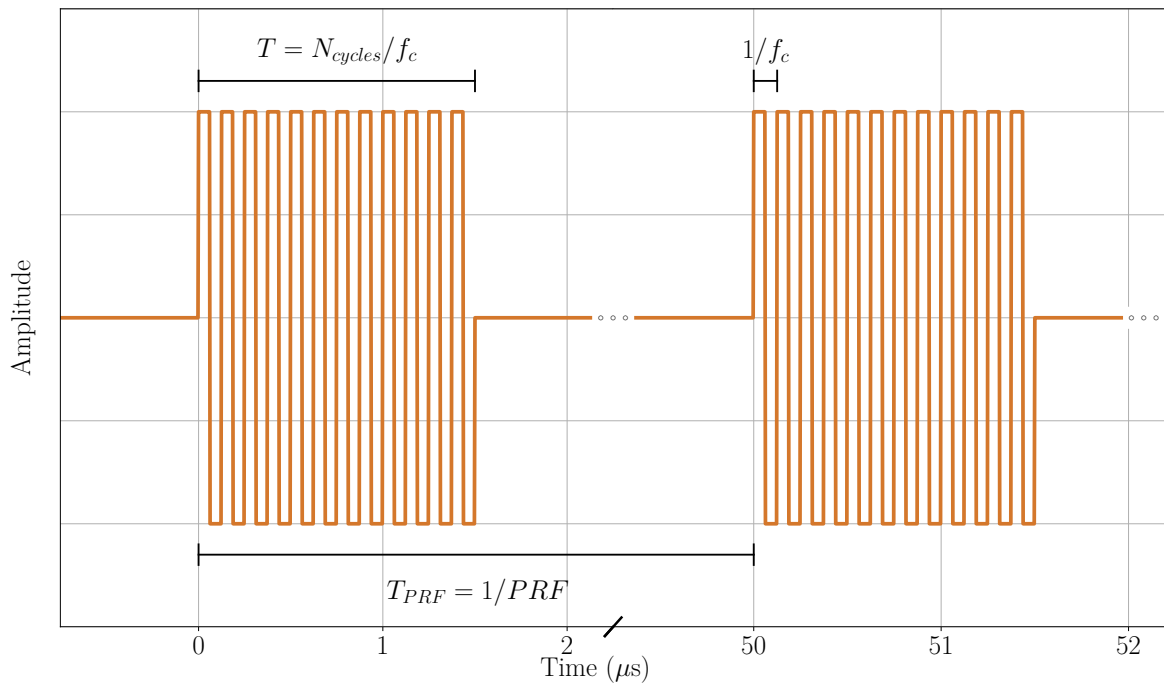


Figure 2.1: Ideal emitted electrical waveform for PW Doppler, with annotations. Note that the time axis is broken to show the pulse repetition period.

2.2.1 Decimation

2.2.2 Clutter Filter

2.3 IMU

2.3.1 Sensor Fusion

Considering the gyro on the Inertial Measurement Unit (IMU) is prone to drift over time, sensor fusion algorithms are employed to combine the measurements from the accelerometer and the gyroscope to obtain a more accurate estimate of the orientation. In our IMU we have access to linear acceleration from the accelerometer, and angular velocity from the gyroscope. There are several ways to perform sensor fusion, including complementary filters, Kalman filters, and more advanced nonlinear filtering techniques.

Complementary Filter

The idea behind the complementary filter is to

Chapter 3

Method

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion